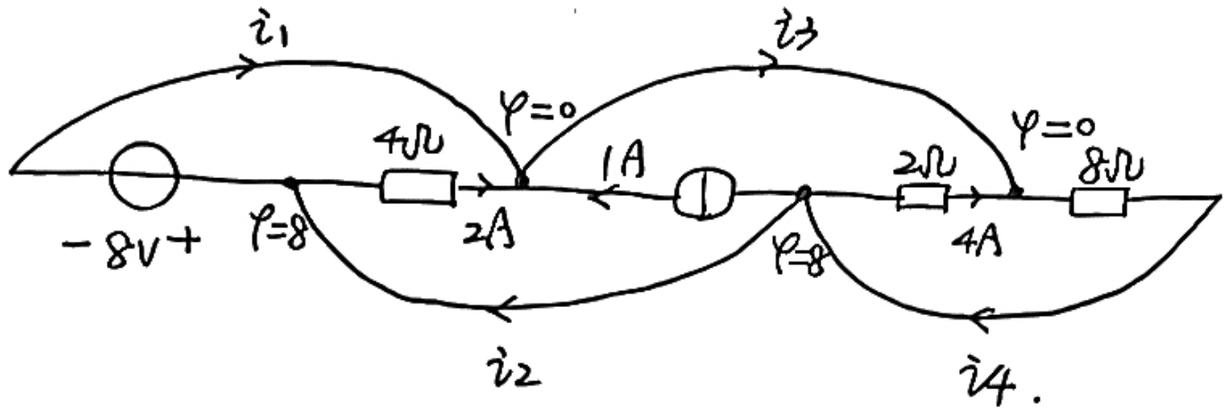
电路样卷3 详解：

1.



标电位法：上例估点，$\varphi=0$，

下例 $\varphi=8V$

标出所有可标明电流，列KCL.

i_4：$8-0=-8i_4$ $\therefore i_4=-1A$.

最右估点：$i_3+4=i_4$ $\therefore i_3=i_4-4=-5A$.

左二估点：$i_1+2+1=i_3$ $\therefore i_1=i_3-3=-8A$.

左一估点：$i_1+2=i_2$ $\therefore i_2=-6A$.

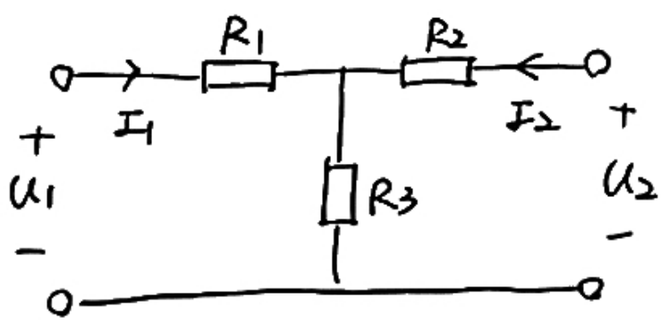
综上：$i_1=-8A$，$i_2=-6A$，$i_3=-5A$，$i_4=-1A$.

2. $I_S=\frac{u_1}{10}+\frac{u_1}{15}=\frac{u_1}{6}$

$u_S=10\cdot\frac{u_1}{6}+u_1=\frac{5}{3}u_1+u_1=\frac{8}{3}u_1$.

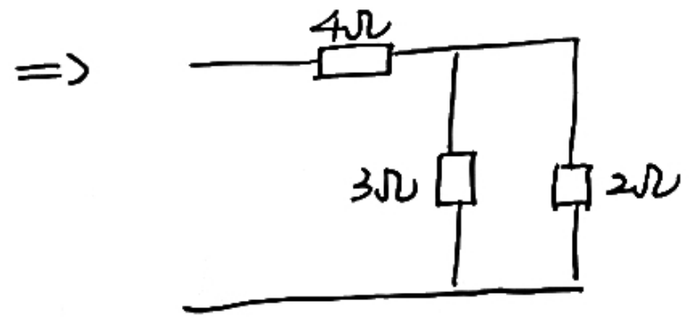
$\therefore R_{ab}=\frac{u_S}{I_S}=\frac{8}{3}\times 6=16\Omega$.

3. 把=端口还原为T型电路，$z_{12}=z_{21}$，网络互易，纯电阻.



$$\begin{cases} u_1=I_1R_1+(I_1+I_2)R_3=(R_1+R_3)I_1+R_3I_2. \\ u_2=I_2R_2+(I_1+I_2)R_3=R_3I_1+(R_2+R_3)I_2. \end{cases}$$

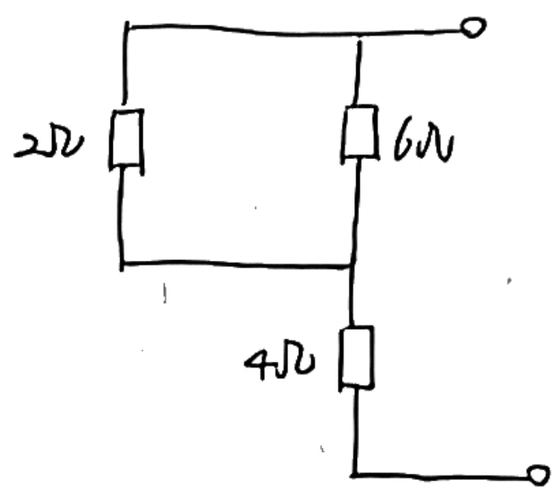
$\Rightarrow R_3=3\Omega$，$R_1=4\Omega$，$R_2=0\Omega$.



$$R_{eq}=\frac{3\times 2}{3+2}+4$$

$$=\frac{6}{5}+4=5.2\Omega.$$

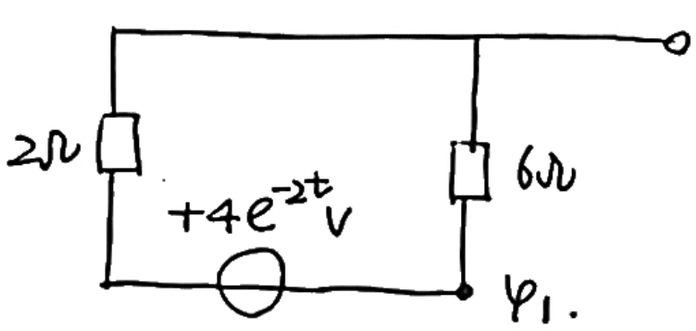
4. 求 R_{eq}：独立源置零：



$$R_{eq} = \frac{2\times6}{2+6} + 4$$

$$= \frac{12}{8} + 4 = 5.5\Omega.$$

求 U_{oc}：由于 2A 电流源串 4Ω 电阻与电压源并联，对外失效.



$$\varphi_a - \varphi_1 = 4\cdot e^{-2t}\cdot\frac{6}{2+6} = 3e^{-2t}(V)$$

对内不失效：4Ω 电阻两端：$\varphi_b - \varphi_1 = 4\times2 = 8V$

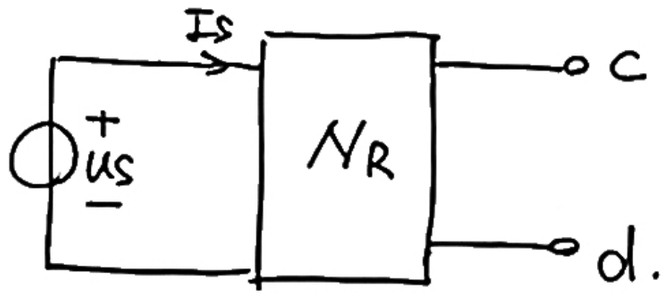
$$\therefore \varphi_a - \varphi_b = (\varphi_a - \varphi_1) - (\varphi_b - \varphi_1)$$

$$= 3e^{-2t} - 8\,(V) = U_S.$$

综上：$U_S = 3e^{-2t} - 8\,(V)$，$R_S = 5.5\Omega$.

5. 注：原题中图 a，b 应当反过来.

求 R_{eq}. 把 b 中独立源置零，加压求流：



发现这正是图 a. $U_S = U_1 = 10V$. $I_S = 2A$.

$$\therefore R_{eq} = \frac{10}{2} = 5\Omega.$$

求 U_{oc}：根据互易定理：$\frac{U_{oc}}{U_2} = \frac{2}{2}$ $\therefore U_{oc} = 5V$.

综上：$U_{oc} = 5V$，$R_{eq} = 5\Omega$.

6. 副边等效至原边：

$$z_L = j\cdot w\cdot \frac{1}{2\pi}\cdot 4\cdot 10^{-3} = j\cdot \frac{2w\cdot 10^{-3}}{\pi}$$

$$z_C = \frac{1}{j\cdot w\cdot \frac{125}{\pi}\cdot 10^{-6}} = -j\cdot \frac{\pi\times 10^6}{125w}$$

谐振： $\frac{\pi\times 10^6}{125w} = \frac{2w\cdot 10^{-3}}{\pi}$

$$\pi^2\cdot 10^6 = 250w^2\cdot 10^{-3} = 25w^2\cdot 10^{-2}.$$

$$w^2 = \frac{\pi^2}{25}\cdot 10^8 \quad \therefore \quad w = \frac{\pi}{5}\cdot 10^4 = 2000\pi = 2\pi f.$$

$$\because f = 1000\,Hz.$$

7. $n^2\cdot 16 = 900 \quad \therefore n = \sqrt{\frac{900}{16}} = \frac{30}{4} = 7.5 \quad \therefore$ 比为 15 : 2

8. 定义式：
$$\begin{cases} u_1 = L_1\frac{di_1}{dt} + M\cdot\frac{di_2}{dt}. \\ u_2 = L_2\cdot\frac{di_2}{dt} + M\cdot\frac{di_1}{dt}. \end{cases}$$

L_2 短路 $\Rightarrow u_2 = 0$.

$$\frac{di_2}{dt} = -\frac{M}{L_2}\cdot\frac{di_1}{dt}.$$

$$= -\frac{3}{4}\cdot\frac{di_1}{dt}.$$

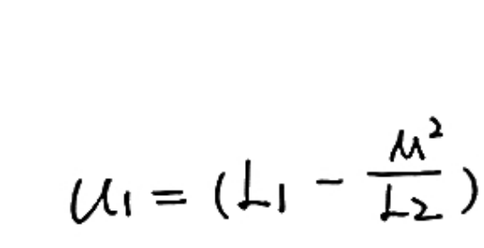
$$\therefore u_1 = (L_1 - \frac{M^2}{L_2})\frac{di_1}{dt}.$$

$$L' = L_1 - \frac{M^2}{L_2} = 6 - \frac{9}{4} = 3.75H.$$

9. 相量图法： Case1： I_A 流入节点、 $\dot{I}_1 + \dot{I}_A = \dot{I}_2$

$\dot{I}_1 = \frac{I_{1m}}{\sqrt{2}}\angle 0°$ ， $\dot{I}_2 = 10\angle 30°$

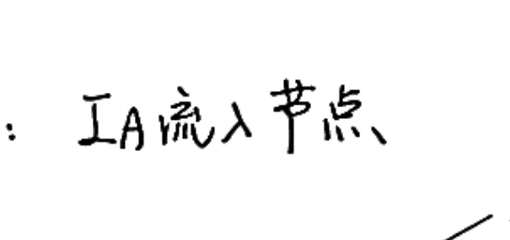
由 $I_A = I_2 = 10$ 的模长关系，仅有一种：



$\therefore I_1 = 10\sqrt{3} = \frac{I_{1m}}{\sqrt{2}} \quad \therefore I_{1m} = 10\sqrt{6}\,A.$

Case2： I_A 流出节点 $\dot{I}_1 = \dot{I}_2 + \dot{I}_A$

同样只有一种.



$I_1 = 10\sqrt{3} = \frac{I_{1m}}{\sqrt{2}} \quad \therefore I_{1m} = 10\sqrt{6}\,A$

综上： $I_{1m} = 10\sqrt{6}\,A$. （答案中0A我认为不妥，题干中说了是正弦电流）

10. A1 测纯抗电流，A2 测纯阻电流，两者正交.

$$I = \sqrt{12^2 + 5^2} = 13A.$$

11. 1)：10Ω 电阻与电流源串联，对外失效 ※※※

节点1：$U_{n1} = 10I_1$.

节点2：$(\frac{1}{20}+\frac{1}{20})U_{n2} - \frac{1}{20}U_{n1} - \frac{1}{20}U_{n3} = 1$.

节点3：$(\frac{1}{40}+\frac{1}{20}+\frac{1}{5+5})U_{n3} - \frac{1}{40}U_{n1} - \frac{1}{20}U_{n2} = \frac{20}{5+5}$.

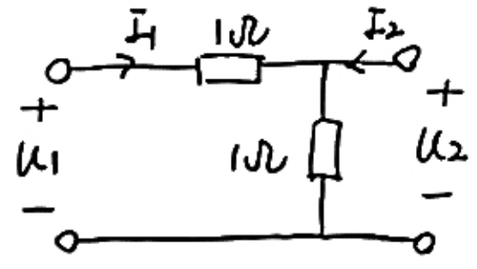
增补：$U_{n3} = 20 - 10I_1$.

2) 解出 $U_{n2} = 20V$，10Ω 电阻电压 $U_1 = 10V$. → 对内讨论不可忽略!!!

$$\therefore U_{1A} = 10V + 20V = 30V$$

三.

端口1：



$U_1 = I_1 + I_1 + I_2 = 2I_1 + I_2$.

$U_2 = I_1 + I_2$.

消去 I_1：得：$U_1 = 2U_2 - I_2$. 再代入第二个方程：

$$I_1 = U_2 - I_2.$$

$$\Rightarrow T_1 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

端口2：$I_1 = 3U_2$

$I_2 = 2U_1 - 4U_2$.

$$\Rightarrow \begin{cases} U_1 = 2U_2 + \frac{1}{2}I_2. \\ I_1 = 3U_2 + 0I_2. \end{cases}$$

$$\Rightarrow T_2 = \begin{bmatrix} 2 & -\frac{1}{2} \\ 3 & 0 \end{bmatrix}$$

$$\therefore T = T_1 T_2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2 & -\frac{1}{2} \\ 3 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & -1 \\ 5 & -\frac{1}{2} \end{bmatrix}.$$

四. 1): $U = \sqrt{12^2+15^2+16^2} = \sqrt{144+225+256} = \sqrt{625} = 25V$

2). ① 直流: $U_R = 12V$

② 基波: L_1与C 并联谐振视为断路, $U_R = 0$.

③ 二次谐波: $L_1 // C$ 串 L_2 发生串联谐振. 视为导线

$$\dot{U}_R = \dot{U}_S = 16\angle 0^\circ \ (V)$$

$$u_{R(t)} = 12 + 16\sqrt{2}\cos(2\omega t) \ (V).$$

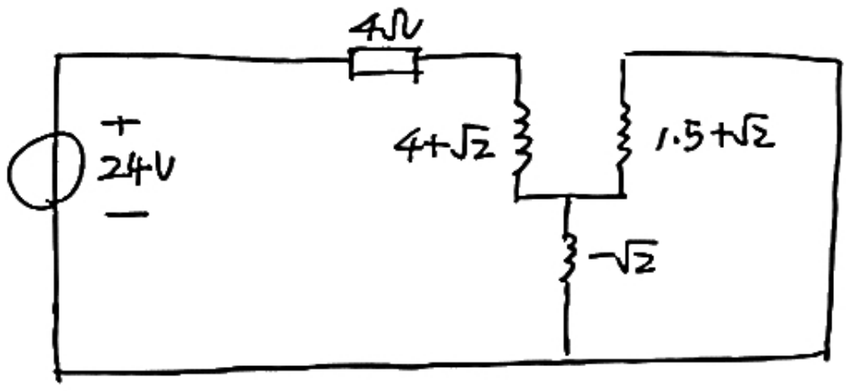
3): 直流: $P_1 = \frac{12^2}{1} = 144W$

二次波: $P_2 = \frac{16^2}{1} = 256W$

$P = P_1 + P_2 = 400W$

(只由电阻产生).

五. $i_L(0^+) = 0$, 去耦:



$$\perp = \frac{(1.5+\sqrt{2})(-\sqrt{2})}{1.5} + 4+\sqrt{2}$$

$$= \frac{-1.5\sqrt{2}-2}{1.5} + 4+\sqrt{2}$$

$$= 4 - \frac{4}{3} = \frac{8}{3}H.$$

$$\tau = \frac{L}{R} = \frac{8}{3}\times\frac{1}{4} = \frac{2}{3}s.$$

$$i_L(\infty) = \frac{24}{4} = 6A$$

$$\therefore \ i_L(t) = i_L(\infty) + [i_L(0^+) - i_L(\infty)]e^{-\frac{t}{\tau}}$$

$$= 6 - 6e^{-1.5t} \ (A)$$